

ALX Foundations: Milestone 5 Worksheet

SECTION A: About You

Step 1: Tech Track Interest

After having a taste of the 5 tech tracks last week, please share your personal ranking of them. (1 being the most appealing, 5 being the least appealing.) For your #1 and #2, state WHY you find these 2 the most appealing. (This does not commit you to anything, you are just stating what you currently feel most drawn to.)

As a reminder, they are:

- AWS
- Salesforce
- Software Engineering
- Data Analytics
- Data Sciences

Step 1: Tech Areas Ranking

1 Data Science: the reason for choosing this is that I profoundly believe that A.I will bring forth a rapid pace in innovation and improve the lives of human beings as a society

2 AWS: Cloud technology enables seamless integration of AI models into cloud-based applications, facilitating rapid development, deployment, and scalability of AI solutions

3 Software Engineering

4 Salesforce

5 Data Analytics

Step 2: Strengths

List your personal strengths. List at least 15 total, and be sure to include at least 2 from each of the 3 categories: Talents/Skills, Knowledge, Professional Qualities. (You have already done a version of this. You are encouraged to approach it fresh and *not* refer back to what you have done before.)

Here are some example strengths from each of the 3 categories.

Example Talents & Skills:

- Strong self-discipline and grit

- Good listener
- Good friend
- Analytical mind
- Intuitive Gardener
- Quick to learn new dance steps

Example Knowledge:

- Football statistics
- History of hip hop
- Organic Chemistry
- Gardening techniques
- Intermediate level Perl and C++ -

Example Professional Qualities:

- Organized
- Reliable
- Good Listener
- Writing good Perl scripts

Step 2: My Strengths (Talents & Skills, Knowledge, & Professional Qualities)
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- | |
|---|
| <ol style="list-style-type: none"> 1. Integrity 2. Self starter 3. Growth mindset 4. Analytical Mindset 5. Attention to detail 6. Quick to learn and apply new concepts 7. Intermediate AI Knowledge 8. Intermediate Python Skills 9. Good Prompt Engineering skills 10. Statistical Knowledge 11. Beginner Google Cloud Skills 12. Problem solving 13. Organized 14. Attentive 15. Finishing work ahead of schedule 16. Passion for technology |
|---|

17. Sustainable agriculture knowledge
18. Passion for African Culture
19. Machine learning Algorithms
20. Data Science Knowledge

Step 3: Areas for Growth

Next, list your areas of growth. These are not simply areas of weakness. They should be specific areas *that you actually want to improve upon*. And include a brief WHY. Note that these do not have to be quantified (that is, they do not need to use a SMART goal format). Please list at least 6 areas for growth.

For example:

- Do more exercise so that I feel stronger and healthier in my body.
- Learn how to better communicate with people from other cultures so that I am more likely to be hired by an international company.
- Visit my mother more often because I care about her and want to make sure she's doing ok.

Step 3: My Areas for Growth

1. Communication: learn how to communicate better so that I can express my knowledge in AI better.
2. Self Discipline: With self discipline I'll practice the daily 3 and get to see tremendous changes.
3. Collaboration: I want to learn how to be part of teams and contribute directly to improvement of companies processes.
4. Continuously learn and practice using new and emerging technologies to grow my skills set.
5. Morning pages: I need to practice morning pages often to improve on my writing and creative skills.
6. Meditation: Meditation will improve my focus and mental health to tackle grand challenges throughout life.
7. Artificial Intelligence know how, with the rise of A.I I aim to equip myself with the knowledge and the skills to create and effectively use A.I systems.
8. Self awareness: I aim to heighten self awareness so that I can experience self growth.

Step 4: GCGOs

Now, list the 3 GCGOs that most strongly resonate with you and for each, write 1-2 sentences about WHY they resonate with you. As a reminder the 14 GCGOs are:

- Urbanization
- Education
- Infrastructure
- Healthcare
- Climate change
- Governance
- Job creation
- Agriculture
- Natural resources
- Arts, culture, and design
- Tourism
- Empowerment of women
- Regional integration
- Wildlife conservation

For example:

- Wildlife conservation because I really love animals and think they have just as much right to be on the planet as humans do. I feel especially passionate about not letting mountain gorillas and rhinos go extinct.
- Education because I believe our knowledge and mindsets are the best assets we have as humans. I want to live in a world where anyone who wants it has access to quality education.
- Agriculture because I've always loved gardening and I want people to have healthy food to eat at low cost. I'm excited when I hear about people finding ways to "hack" agriculture to produce greater crop yields and lower costs.

Step 4: My Top 3 GCGOs

1. Urbanization: Traffic congestion in Nairobi matters because it hampers productivity, increases stress levels, and reduces the overall quality of life for residents. Additionally, it contributes to environmental pollution and hinders the city's economic growth and development.
2. Agriculture: African nations need to embark on transforming agriculture with new tech practices . Sufficient food reserves will ensure that other areas of the economy are focused on and effectively developed as well.

3. Wildlife and Biodiversity conservation: Human beings have been on a rampage invading and taking over spaces that are meant for wildlife resulting in more conflicts. By focusing on improving gradually our environment through planting trees, we can create more biodiversity and lessen encroachment on Wildlife reserves

Step 5: Areas of Care

In addition to the GCGOs, list at least 6 other things you deeply care about, including things you want.

For example:

- My 2 sisters. I want to make sure they are safe and can keep going to school.
- The neighbor who lives down the road who is always kind to me and is sick now.
- I want to buy some land and build a house.
- I care about wild animals and am passionate about saving endangered species from extinction.
- I would like to have a healthier relationship with my father.

Step 5: What I deeply care about

1. Agriculture is one of my biggest passions: Africans shouldnt be dying from famine in 2023. We need to improve our systems to ensure that this is managed.
2. I am passionate about Artificial Intelligence: A.I will greatly improve our economies, governance, innovation as well as productivity in existing industries
3. I care about not only the advocacy of protecting our environments but actually taking part in Exercises such as tree planting. This is the simplest way to tackle climate change instead of just talking about it in conferences where keynote speakers get to using Jets
4. I care deeply about our wildlife. We need to protect the endangered species and put an end to poaching through real time monitoring of our parks.
5. I would love to improve communication both in professional settings and in my personal life. This will keep me in touch with the people I care about as well as help me network.
6. I have a deep passion in improving myself so that I can be a beacon of change among my peers and also in my nation Kenya
7. I want more people to be concerned about electing good leaders as they directly have an impact on their quality of life .
8. I would like to involve the relevant authorities in implementing systems to trap plastic

from getting into our waterways hence making water safe to use.

Based on

SECTION B: Problems

Step 6: Identifying Local Problems

Look around in your local community (for now, we'll define your *community* as a group of people you care about and/or a certain geographical area that you live in or near). What are some problems that you see in your local community? List at least 10.

For example:

- There is a lot of traffic that makes it hard to get anywhere in a reasonable amount of time.
- There is a lot of trash on the street and it is unsanitary.
- There is not enough water.
- Electricity is intermittent and inconsistent.
- People don't have regular and consistent internet/network access.
- Kids are leaving school too early in order to go to work.
- My church group doesn't have enough funds to care for the people that need help.
- Many people on motorcycles are not wearing helmets.
- There are not enough doctors to serve all the people who need medical help.
- There are a lot of car accidents.

Step 6: Problems in my community

1. Nairobi faces a lot of traffic throughout the day and this is a problem that should be addressed to ensure that the city infrastructure grows and accommodates the growing population
2. There's not enough clean water
3. Blocked sewerage systems
4. Not enough green spaces in the city
5. Poor public transport systems
6. Boda Boda riders using pedestrian lanes
7. Boda Boda riders break traffic rules
8. Poor road networks

9. Expensive private health care
10. Poor drainage facilities that lead to flooding during rainy seasons
11. Youth unemployment.

Steps 7-9: Your Chosen Problem

As a next step, we would like you to narrow in on 1 problem facing your community that feels most compelling to you and is related to one of your top 3 GCGOs. It can be the same, similar, or different as the problem you focused on for Milestone 4. Please choose a problem that is compelling to you personally. **This problem will be what you bring to your team discussion the first week of Month 2.**

Step 7: My chosen problem is:

Nairobi faces a lot of traffic throughout the day and this is a problem that should be addressed to ensure that the city infrastructure grows and accommodates the growing population.

Step 8: I care about this problem because:

With my knowledge in the efficiency of high-speed trains and computer vision, I care about the problem of traffic because I understand the potential of these technologies to revolutionize transportation systems. By addressing traffic congestion in Nairobi through the implementation of high-speed trains and intelligent traffic management, I can contribute to improving the efficiency, sustainability, and overall quality of life in the city.

Step 9: My chosen problem is related to the following GCGO:

Infrastructure

Step 10: Vision statement

Next, craft a vision statement for this problem. What do you imagine your community would be like when your chosen problem has been solved?

Step 10: My vision statement

In a future where traffic congestion is effectively addressed through the implementation of high-speed trains and intelligent traffic management, our community in Nairobi thrives with seamless and efficient transportation. The city is characterized by smooth traffic flow, reduced travel times, enhanced connectivity, and improved air quality. Commuters enjoy stress-free journeys, productive use of their time, and a sustainable transportation system that supports the city's growth and prosperity.

Step 11: Mission statement

As the last part of this section, we'd like you to update your personal mission statement. Revisit this from Week 2 and revise/restate as you wish. It is normal and expected that your statement will have evolved since Week 2. How will you personally move toward the world you want to create? (If you need a refresher, you can find it [here](#).)

Make sure you:

- State what your personal mission is.
- State one or more of your key strengths.
- State one or more areas that you pledge to further develop.
- State one or more of your core values.

Step 11: My mission statement

My personal mission is to leverage technology, specifically AI, to drive positive change and create a more equitable and sustainable world. I am committed to using my boundless curiosity and passion for innovation to address societal challenges and improve the lives of individuals and communities.

One of my key strengths lies in my ability to adapt and embrace new concepts. I thrive in dynamic environments and have a strong capacity for continuous learning. This enables me to stay at the forefront of technological advancements and apply them effectively to create meaningful solutions.

I pledge to further develop my leadership skills and cultivate empathy and inclusivity. By honing my ability to inspire and empower others, I aim to foster collaborative environments that promote diversity and amplify voices that have been historically marginalized.

Transparency, integrity, and accountability are among my core values as a leader. I believe in upholding the highest ethical standards and leading by example. By fostering a culture of trust and open communication, I strive to create an environment where diverse perspectives are respected, and collaboration thrives.

To move toward the world I want to create, I will actively engage in ongoing education and professional development to deepen my knowledge and expertise in AI, machine learning, and related fields. I will seek out opportunities to collaborate with like-minded individuals and organizations to drive meaningful impact. Additionally, I will advocate for responsible and ethical use of technology, ensuring that its benefits are accessible to all and that its potential risks are addressed proactively. By continuously striving to align my actions with my personal mission, I will actively contribute to the creation of a more equitable, inclusive, and sustainable future.



**Please go back to Canvas and continue with your learning content.
You will be prompted on when to return to complete Section C.**

SECTION C: Users/Market Inquiry

Steps 12-13: Your Market

In Month 2, you will learn more about specific people impacted by your chosen problem. In business terms, this can be called your *market*. This term implies that you are creating a solution— a product or service— that will be given to or sold to this particular audience.

For now, you can start thinking more concretely about who exactly this problem impacts. That is, who are you solving this problem for? What criteria does a person meeting to be included in your definition of a user/part of your market? These could include demographic information, psychographic information, and behavioral traits. Where do they live? Are there age and/or gender requirements? Socio-economic requirements? List at least 4.

Step 12: Define the criteria for being a user/ part of your market:

1. Location: Individuals residing in Nairobi, Kenya or commuting within the city.
2. Commuters: Individuals who regularly travel within the city for work, education, or other purposes.
3. Age and Gender: No specific age or gender requirements, as the problem affects people of various age groups and genders.
4. Socio-Economic Background: People from diverse socio-economic backgrounds who encounter traffic congestion as part of their daily routine, including both low-income and high-income individuals.
5. Commuting Frequency: Individuals who commute frequently within Nairobi, such as daily commuters or regular travelers.
6. Travel Distance: Individuals who travel both short and long distances within Nairobi, experiencing traffic congestion during their journeys.
7. Occupation: Professionals, students, or individuals whose daily activities are affected by traffic, including those traveling to work, schools, universities, or other educational institutions.
8. Environmental Consciousness: Individuals who prioritize sustainable transportation options and are interested in reducing their carbon footprint by using efficient transportation alternatives.
9. Technological Proficiency: Users who are comfortable using technology and digital platforms, as potential solutions may involve the use of apps, real-time updates, and digital navigation systems.
10. Decision-making Authority: Individuals who have decision-making power or influence in implementing transportation-related changes, such as government officials, urban planners, or policymakers.

Next, estimate the total number of people in your identified community.

Step 13: Estimated community/market size- Your TOTAL ADDRESSABLE MARKET (TAM)

List estimated size and how you arrived at this number:

The total addressable market (TAM) of Nairobi population that can be affected by traffic congestion is estimated to be around 6.5 million people. This is based on the 2020 census data, which shows that the population of Nairobi is around 4.4 million people. However, the city's metropolitan area is home to an additional 2.1 million people, who are also likely to be affected by traffic congestion.

70% of the Nairobi population regularly experiences traffic congestion during their daily commutes. This percentage can be adjusted based on more specific data and research findings.

Using these assumptions, we can estimate the total addressable market (TAM) for solutions targeting traffic congestion in Nairobi as follows:

TAM = Population of Nairobi * Percentage of Population Affected by Traffic Congestion

TAM = 6.5 million * 70%

TAM = 6.5 million * 70% = 4.55 million

Step 14: User Persona

Now create your user persona. This should be a fictitious composite who represents the average “user”, or member of your community affected by your problem.

Step 14: User Persona

- Their name (this made up): John Mulei
- Gender: Male
- Age: 35
- Profession: Software Engineer
- Education history: Bachelor's degree in Computer Science from the University of Nairobi
- Annual income: 2,000,000
- Access to technology: Has a smartphone and a laptop computer
- Other relevant information: John lives in South C and commutes to work in the city center every day. His commute takes him an hour and a half each way, and he often gets stuck in traffic. John is frustrated by the traffic congestion, and he believes that it is a waste of his time and money. He is also concerned about the environmental impact of traffic congestion.

Step 15: Potential Interviewees

Think about specific people who are part of this market. These should be real people you know about or know directly, who you could potentially conduct user interviews with to better understand how your problem affects them. Please list at least 4 people.

Step 15: People I could interview:

1. Name: Sarah Kimani

Occupation: call center agent

Age: 25

Gender: Female

Relationship to Problem: Sarah works in Nairobi upperhill and relies on smooth traffic flow for timely deliveries and customer satisfaction. She is affected by traffic congestion, which leads to her sometimes getting late to work.

2. Name: John Mwangi

Occupation: Commuter

Age: 28

Gender: Male

Relationship to Problem: John works in the city center and commutes daily from a suburban area in Nairobi. He faces long commuting hours and is interested in efficient transportation solutions to save time and reduce stress.

3. Name: Grace Achieng

Occupation: Receptionist

Age: 33

Gender: Female

Relationship to Problem: Grace works as a receptionist in Nairobi and deals with the challenges of traffic congestion on a daily basis. She can provide insights into the impact of congestion and the effectiveness of current systems.

4. Name: Michael Kamau

Occupation: Driver Matatu

Age: 45

Gender: Male

Relationship to Problem: Michael operates a matatu (public minibus) in Nairobi and experiences firsthand the effects of traffic congestion on the efficiency of public transportation. He can share his perspective on the challenges faced by drivers and the impact on passenger experience.

Step 16: User Interview Questions

In Month 2, you will need to ask your interviewees questions that help you better understand how your chosen problem impacts them. Please write out at least 6 questions that you could ask. An example question is, "Tell me more about how lack of consistent electricity impacts your business." Another example question is: "Have you or someone in your family needed medical attention in the past 12 months and been unable to receive adequate care?"

Step 16: User Interview Questions

1. How does traffic congestion in Nairobi impact your daily routine or activities?
2. Can you describe a specific instance when traffic congestion caused inconvenience or delays for you?
3. What are the main challenges you face when it comes to commuting or traveling within Nairobi?
4. How does traffic congestion affect your productivity or efficiency at work?
5. Have you ever missed important appointments or events due to traffic congestion? If yes, can you share your experience?
6. Are there any specific areas or routes in Nairobi where you consistently encounter heavy traffic? How does this affect your overall experience?
7. What measures or solutions do you think could help alleviate traffic congestion in Nairobi?
8. In your opinion, how do you think an efficient high-speed rail system or intelligent traffic management through computer vision could improve the traffic situation in Nairobi?
9. How does traffic congestion impact your overall quality of life or well-being?
10. Can you share any stories or experiences related to traffic congestion in Nairobi that have had a significant impact on you personally or professionally?

PART D: Hub Activities Report

Step 17: Hub Activity #1 - Peer Coaching

Please report on your peer coaching session. (Instructions [here](#).)

Step 17: Hub Activity #1 Report

A: The first and last name of the peer you had your peer coaching session with:

Sandra Wanyama

B: The date AND time that the coaching meeting took place:

Friday 10 am - 1 pm at The foundry.

C: Did you feel any anxiety before or during the meeting? Why or why not?

No, I was eager to meet my peer coaching partner and get to know their progress from week one. This short term growth would help me stay motivated.

D: How valuable was the session with your peer? Explain.

During the peer coaching session, I had a positive experience and gained valuable insights into my coaching partner's time management strategies, particularly their use of the daily 3 to maintain focus. It was inspiring to witness their dedication and progress towards their goals.

E: Did you make a commitment to your peer? If yes, what was the commitment that you made?

Starting from 09/06/2023, I commit to regularly checking in with them to track their progress in developing their people skills and completing the ALX modules. I will make weekly calls every Friday to provide support and ensure their continuous improvement.

F: Did you find anything surprising and/or gain any new insight due to the meeting?

I will improve on my time management skills further and also build on my daily habits.

G: How helpful did you find the coaching session overall? Explain and share your experience honestly.

The coaching sessions have proven to be invaluable, providing me with valuable insights to enhance my skills. Moving forward, I am committed to developing stronger collaborative skills and fostering effective teamwork with my peers.

Step 18: Hub Activity #2 - User Interviews

Please report on the 2 interviews that you conducted at your City Hub user interviews. (Instructions [here](#).)

Step 18: Hub Activity #2 Report

A. The date and time that you visited your City Hub:

Thursday 08/06/2023.

Friday 09/06/2023.

10 am - 1 pm.

B. The full names of the two peers you interviewed:

AnnJulie Kang'ethe.

Sandra Wanyama.

C. In 2 - 3 sentences, share your experience of the entire interview process.

The interview process yielded valuable and profound insights into the daily commuting experiences of Nairobi residents. Through the use of open-ended questions, I successfully elicited detailed and thoughtful responses from my peers, enriching my understanding of the problem and potential solutions.

D. In 2 - 3 sentences, share your experience of the entire feedback process.

Following the completion of the interview, my peers provided feedback indicating that the overall process was effective and unhurried. They specifically highlighted the well-worded, open-ended, and unbiased nature of the questions, which contributed to a successful and insightful interview experience.

Once you have completed this worksheet, export/convert to .pdf, rename it per the instructions, and upload to Canvas as your Milestone 5 Submission. Celebrate a job well done!